Teaching Dossier

William Duan, Vancouver School of Economics, University of British Columbia

November 2026

This dossier sets out my teaching record and the evidence for it. An online version, with every evaluation report and further course materials, is at https://william96123.github.io/teaching/.

Contents

		Page
1	Teaching statement	2
2	Teaching responsibilities	4
3	Evidence of teaching effectiveness	5
4	Teaching innovations	5
5	Contributions beyond my own sections	6
6	Courses I am prepared to teach	6
	Guide to the appendices	6
A	PhD Math Camp course outline (sample syllabus)	7
B	Proof and real analysis problem set, PhD Math Camp	9
C	ECON 600 tutorial outline	12
D	GARP coding guide, ECON 600 (sections 1 to 6)	14
E	ECON 600 tutorial notes, choice under risk	19
F	ECON 101 discussion worksheet, Week 2	26
G	Evaluation report with all student comments, ECON 600, Fall 2025	28

Teaching Statement

William Duan — Vancouver School of Economics, University of British Columbia

Over seven years as a teaching assistant I have taught economics at every level UBC offers, from first-year principles to the PhD microeconomic theory sequence (ECON 600 and 601), which I have taught every year since 2021. In Summer 2026 I was the instructor for the department's MA and PhD Math Camp, developing material the department intends to build future camps on. Teaching the PhD core, the most technically demanding material in the curriculum, alongside first-year and third-year sections has taught me to present material at the level each audience requires.

I am prepared to teach any course in the undergraduate curriculum, including macroeconomics and econometrics. My strongest interest and deepest preparation are in microeconomics, from principles through game theory, information economics, and auctions and market design, and, at the graduate level, in mathematics for economists and the microeconomic theory core. I find teaching deeply rewarding. Introducing students to economics is among the parts of academic work I value most, and I intend for it to remain central to my career.

Teaching philosophy

At a university, students lead their own learning. Understanding is something they build through their own work, and no lecture can do that work for them. The instructor's role is to make that work possible, and I see it as three tasks.

The first is to give students a reason to do the work. Effort that students choose to make depends on seeing why the material matters, so motivation is where teaching starts rather than an addition to it. The second is to set out clearly what students are expected to understand and to what level. Students leading their own learning need to know where they are going, and a clear target lets them judge their own progress instead of waiting for an exam to judge it. The third is to help them past the points where they get stuck. These are the moments when self-directed learning is most likely to stall, and where an instructor adds the most.

The three tasks depend on one another. Motivation without a clear target produces effort that drifts. A clear target without support leaves students facing the hardest parts alone. Support is only useful once students are invested enough to want it and know enough about the goal to see where they are stuck.

Teaching practice

I carry out these tasks mostly through problems. A tool is easier to motivate when students first see the problem it was built to solve, whether an academic question or a practical one, so in my sections the material is introduced through questions, mostly from problem sets and past exams. We solve them together, with me calling on students as we go. Working a problem also tells students where they stand. A problem they can finish confirms what they understand, and the point where they get stuck shows them, and me, which part of the general idea they have not yet grasped. Handing students the reasoning in this way does not depend on a small room, and the same principle has been applied successfully in large introductory economics courses.[1]

Proofs are a core part of theory courses. Intuition comes first and the technical steps second, because the aim of a proof is not to take students through each step but to give them enough understanding of the result that they feel able to use it. I present a result in four stages. I begin with what it says and where it applies, either by showing what would go wrong without it, so that students see what it gives them, or through a graphical or numerical illustration where one exists. I then explain why the result is not obvious, which identifies the

[1] Denny, E. (2014), "Reinterpreting the economics textbook: A student assignment," *International Review of Economics Education* 16, 100–110.

core fact the proof has to establish. Next I go through the intuition of the proof, built on those first two stages. Only then do I turn to the technical details and the formal argument. The same approach applies to abstract analytical results at any level.

In my tutorials I present one clean, linear solution first and save alternative routes for the end. A clean solution shows students how an argument should go, and in the PhD core I also want them to recognize when one does not. For some time I have presented proofs that are correct in their approach but omit a few necessary and subtle steps, and asked students to find where the argument is incomplete. This year I have started generating flawed proofs with weaker AI models for the same exercise. Reading a plausible argument and finding where it fails is among the most valuable skills a graduate student can take from a theory course, and lectures do not train it.

Finally, helping students past the hard parts depends on knowing where those parts are, and students know far more about what they do not understand than an instructor can observe. Much of what I do outside the classroom is meant to bring that knowledge out. In a class of twenty I work at making students comfortable with me, through a casual manner and an interest in their lives beyond the course, so that they will bring me even their small questions. When they do, I try to reconstruct how they were thinking before I say anything, because the useful answer is the correct solution attached to the place where their reasoning went wrong. In a large class the useful connection is among the students. In ECON 101 I put the whole section into one group chat and routed every question through it. Most were answered by classmates before I saw them, and my job became confirming the answer or adding a small correction. Students who would never have emailed a teaching assistant asked freely there.

Teaching large classes

Most of my teaching has been in sections of about twenty, and not every practice that works at that size survives in a lecture hall. The question channel from ECON 101 is the one I am most confident will scale, because I have already run it in a larger class. In a large ECON 101 or intermediate microeconomics (ECON 301) course I would route all questions through one moderated forum,[2] with a small amount of bonus credit for correct answers to classmates. This makes the strongest students a resource for the whole class without assigned groups, whose composition is hard to get right and which students rarely welcome. Class time would remain devoted to lectures and worked problems, and I would keep calling on students as a problem is solved, so that a large room follows the reasoning step by step rather than receiving only the answer.

Evidence of teaching effectiveness

Course	Years	Sections	Responses	Median, /5	Good or very good
ECON 600/601 (PhD micro theory)	2021–2026	10	104	4.8–5.0	100%
ECON 514 (information and incentives)	2021	1	7	4.8	100%
ECON 101 (principles of micro)	2020, 2025	3	34	4.5–4.9	92–100%
ECON 303, 355 (intermediate micro, trade)	2019–2020	2	44	4.5–4.7	70–74%

Across the sixteen reports, an average of 96% of responding students rated my overall contribution good or very good. I received the Vancouver School of Economics Agustín Beltrán Award for Excellence in Teaching, the department's award for teaching assistants, in 2024, 2025 and 2026. My teaching dossier gives the full results, sample materials and my other contributions to teaching, and is also online at william96123.github.io/teaching.

[2] Barile, L., and A. Ghosh (2026), "Understanding online peer-assisted learning with forums in the 'digital era'," *Journal of Economic Education*, published online 29 July 2026.

2. Teaching responsibilities

Course	Level	Role	Terms	Class size
Math Camp	MA and PhD	Instructor	Summer 2026	14 PhD, plus MA
ECON 600, micro theory I	PhD	TA	Fall 2021 to 2025	18 to 25
ECON 601, micro theory II	PhD	TA	Winter 2022 to 2026	17 to 23
ECON 514, information	MA	TA	Winter 2021	16
ECON 303, intermediate micro	Undergraduate	TA	Winter 2020	53
ECON 101, principles of micro	Undergraduate	TA	Fall 2020, Summer 2025	41 to 60
ECON 355, international trade	Undergraduate	TA	Fall 2019	100

Math Camp. In Summer 2026 I taught the department's MA and PhD math camps under the supervision of Prof. Jesse Perla. I wrote the proof and real analysis lecture and its problem set in full (Appendix B). For the other PhD lectures I extended the MA slides with PhD material and added a few PhD-level questions to each problem set. The MA sessions ran on recorded faculty lectures, and my part there was the problem sessions and Q&A. Appendix A is the course outline.

PhD microeconomic theory. I have been the teaching assistant for the first-year PhD sequence, ECON 600 and 601, every year since 2021, and I continue in ECON 600 in Fall 2026. Each week I run a ninety-minute tutorial for about twenty students, mostly problem solving at the board, and I hold office hours and grade.

Undergraduate and MA courses. Between 2019 and 2025 I was also a teaching assistant for ECON 355, 303, 514 and 101, in classes of 16 to 100 students. In ECON 101 I led two online discussion sections in Fall 2020 and returned to the course in Summer 2025.

3. Evidence of teaching effectiveness

At the end of each term UBC asks students to rate the teaching assistant's overall contribution to the course on a five-point scale, from very poor to very good. The table lists every term in which I have been evaluated, with UBC's median rating and the share of responding students who rated my contribution good or very good. Across the sixteen reports that share averages 96%.

Term	Course	Enrolled	Responses	Median (of 5)	Good or very good
Fall 2019	ECON 355	100	34	4.5	74%
Winter 2020	ECON 303	53	10	4.7	70%
Fall 2020	ECON 101 (L5C)	48	17	4.9	94%
Fall 2020	ECON 101 (L5D)	60	13	4.9	92%
Winter 2021	ECON 514	16	7	4.8	100%
Fall 2021	ECON 600	25	13	4.9	100%
Winter 2022	ECON 601	23	8	4.9	100%
Fall 2022	ECON 600	21	16	5.0	100%
Winter 2023	ECON 601	20	12	5.0	100%
Fall 2023	ECON 600	21	8	4.8	100%
Winter 2024	ECON 601	22	7	5.0	100%
Fall 2024	ECON 600	18	12	5.0	100%
Winter 2025	ECON 601	17	6	4.9	100%
Summer 2025	ECON 101	41	4	4.5	100%
Fall 2025	ECON 600	22	16	5.0	100%
Winter 2026	ECON 601	23	6	4.8	100%

The two lowest shares, 74% and 70%, come from my first two terms as a teaching assistant, in ECON 355 and ECON 303. Every term since has been at 92% or above. The PhD classes have about twenty students, so each of those terms rests on 6 to 16 responses.

Appendix G reproduces UBC's full report for ECON 600 in Fall 2025, including every student comment from that term. All sixteen reports, with their comments, are in the online dossier.

Award. I received the Agustín Beltrán Award for Excellence in Teaching from the Vancouver School of Economics in 2024, 2025 and 2026. The award goes to two teaching assistants each year.

4. Teaching innovations

AI-assisted coding. ECON 600 includes an optional coding exercise in which students test experimental choice data for violations of the generalized axiom of revealed preference. I wrote the step-by-step guide for it (Appendix D). This year I rebuilt the exercise around AI-assisted coding. Students may use AI as much as they like, but only a lighter model, which forces them to explain the algorithm clearly. Their effort goes into the economics of the test and into checking the code rather than into syntax.

Finding the gap in a proof. For some time I have given ECON 600 students proofs that are correct in approach but omit a few necessary steps, and asked them to find where the argument is incomplete. This term I started generating flawed proofs with weaker AI models, which also teaches students to read AI output critically. There are no evaluations from this term yet.

A class-wide question channel. In ECON 101 I put the whole section into one group chat and routed every question through it. Most questions were answered by classmates before I saw them, and my job became confirming the answer or adding a small correction. Students who would never have emailed a teaching assistant asked freely there. My teaching statement describes how I would carry this into a large lecture course.

5. Contributions beyond my own sections

Math Camp material. The department set out in 2026 to develop Math Camp material that future camps could build on, and entrusted me with both the MA and PhD material under Prof. Perla's supervision. For the MA camp I organized and augmented existing material. For the PhD camp I wrote the proof and real analysis module in full and extended the other modules with PhD-level material.

A record for future teaching assistants. This year I am writing notes after each ECON 600 tutorial so that future teaching assistants inherit a full record of the course. Appendix E is one session. I have also prepared tutorial outlines for ECON 600 and 601 that set out the objectives and a week-by-week schedule for each (Appendix C).

ECON 601 problem sets. Over the last two years I co-designed problem sets for ECON 601.

Junior PhD students. I also have informal research conversations with first- and second-year PhD students about their early work.

6. Courses I am prepared to teach

Undergraduate. Principles and intermediate microeconomics, principles and intermediate macroeconomics, and introductory econometrics. Game theory, information economics, and auctions and market design.

Graduate. Microeconomic theory, game theory and information economics, and mathematics for economists.

Guide to the appendices

A. PhD Math Camp course outline. The camp did not distribute a formal syllabus, so this outline is written from the course materials. It gives the camp's purpose, objectives, schedule and my role.

B. Proof and real analysis problem set. The problem set for the module I wrote in full, reproduced from the public course site.

C. ECON 600 tutorial outline. A baseline plan for the weekly tutorials, written from my notes on each session. It covers the tutorials only, not the lectures.

D. GARP coding guide. The student guide to the ECON 600 coding exercise, sections 1 to 6. The extensions section and an answer check are omitted here, and the full guide is in the online dossier.

E. ECON 600 tutorial notes. My notes on the Fall 2026 tutorial on choice under risk, written for future teaching assistants rather than for students. Material tied to the current problem set and past answer keys has been removed from this version.

F. ECON 101 discussion worksheet. One of the worksheets I wrote for my Fall 2020 discussion sections. Other weeks used materials prepared by the instructor or another teaching assistant.

G. Evaluation report. UBC's complete end-of-term report for ECON 600 in Fall 2025, with all student comments.

PhD Math Camp, course outline

Vancouver School of Economics, University of British Columbia, Summer 2026

William Duan, instructor (under the supervision of Prof. Jesse Perla)

The camp did not distribute a formal syllabus. This outline is written from the course materials, which are public at https://ubcecon.github.io/math_camps/.

Purpose

The camp prepares incoming PhD students for the first-year core sequence. It reviews the linear algebra, optimization and probability that the core courses assume, and it opens with the logic of proof and the real analysis that microeconomic theory relies on from its first week.

Objectives

By the end of the camp, students should be able to do the following.

1. Read a theorem, separating its premises from its conclusion, and write short proofs by direct argument, contrapositive and contradiction.
2. Work with open, closed and compact sets, sequences and limits, and continuous functions in $\mathbb{R}^n$, the language in which existence results are stated.
3. Treat matrix products as linear combinations, and use inverses to analyse systems of linear equations.
4. Solve unconstrained and equality-constrained optimization problems, check second-order conditions, and interpret a Lagrange multiplier.
5. Use conditional probability, Bayes' rule, moments and conditional expectation with confidence.

Schedule

The camp ran as four modules, in the order below. Each module combined a lecture with a problem session built on its problem set.

Module 1 · Proof and real analysis. Statements, connectives and implication, and three strategies for proving $P \Rightarrow Q$. Quantifiers and biconditionals. Sets, open and closed sets, completeness of the real line, and bounded and compact sets. Sequences, monotone sequences, subsequences and limit points. Continuity and the main theorems on continuous functions.

Module 2 · Linear algebra. Vectors, inner products and norms. Orthogonality, linear independence and span. Matrix operations and their properties, matrix-vector products read as dot products and as linear combinations. The identity matrix, inverses, and systems of equations.

Module 3 · Optimization. A calculus review, then concavity and its derivative characterization. First- and second-order conditions and uniqueness in unconstrained problems. Equality-constrained problems, the Lagrangian, and the interpretation of the multiplier, worked through utility maximization.

Module 4 · Probability. Probability spaces and the axioms. Conditional probability, the law of total probability and Bayes' rule. Random variables, distribution and density functions, moments and quantiles. Multivariate distributions, conditional expectation, the law of iterated expectations, and standard inequalities.

Materials and assessment

Each module has a slide deck and a problem set with full written solutions, 73 questions in all. Questions that go beyond the MA level are marked as the PhD track. The problem sets were for practice and self-assessment, and the solutions were released to students.

My role

I designed the PhD sessions of the camp with Prof. Perla. I wrote the proof and real analysis lecture and its problem set in full. For the other three modules I extended the existing faculty slides with PhD-level material and added PhD-level questions to each problem set. The department intends to keep this material as the basis for future Math Camps.

Proof and Real Analysis — Practice Problems

William Duan

These practice questions accompany the Proof and Real Analysis lecture. Each asks you to prove a result stated in the lecture. A few of the harder proofs are marked *(Challenging)* — attempt the others first.

Question 1

Prove De Morgan's laws for sets: for $A, B \subseteq X$,

$$(A \cup B)^c = A^c \cap B^c \qquad \text{and} \qquad (A \cap B)^c = A^c \cup B^c.$$

Question 2

Prove the following stability properties of open and closed sets in $\mathbb{R}^n$.

a. The union of an arbitrary family of open sets is open.
b. The intersection of finitely many open sets is open.
c. Using De Morgan's laws, deduce that an arbitrary intersection of closed sets is closed, and a finite union of closed sets is closed.

Question 3

Let $A \subseteq \mathbb{R}$ be non-empty and let $s \in \mathbb{R}$ be an upper bound of A. Prove that $s = \sup A$ if and only if

$$\forall\, \varepsilon > 0,\ \exists\, a \in A \text{ such that } a > s - \varepsilon.$$

Question 4

Using the definition **compact = closed and bounded** (in $\mathbb{R}^n$), justify each claim.

a. $[0,1]^n$ is compact.
b. $(0,1)$ is not compact.
c. $\mathbb{R}$ is not compact.
d. The budget set $\{x \in \mathbb{R}^n : x \geq 0,\ p \cdot x \leq w\}$ is compact, where $p \gg 0$ (all prices strictly positive) and $w > 0$.

Question 5

Prove that if a sequence (x_k) in $\mathbb{R}^n$ converges, then its limit is unique.

Question 6

Let $x_k \to x$ and $y_k \to y$ in $\mathbb{R}$. Prove:

a. **(Algebraic Limit Theorem)** $x_k + y_k \to x + y$; $c\,x_k \to cx$; $x_k y_k \to xy$; and, if $y \neq 0$ and $y_k \neq 0$, $x_k/y_k \to x/y$.
b. **(Order Limit Theorem)** if $x_k \leq y_k$ for all k, then $x \leq y$.
c. **(Squeeze Theorem)** if $x_k \leq z_k \leq y_k$ for all k and $x_k, y_k \to L$, then $z_k \to L$.

The quotient limit in (a) is the most delicate part.

Question 7

a. Prove the **Monotone Convergence Theorem**: every bounded, monotone sequence in $\mathbb{R}$ converges. (Prove the non-decreasing case; the non-increasing case is symmetric.)
b. Show **directly** from the ε-K definition — without invoking the MCT — that $x_k = 1 - \frac{1}{k}$ converges to 1.

Question 8

a. Prove the **sequential characterisation of closed sets**: $F \subseteq \mathbb{R}^n$ is closed if and only if every convergent sequence $(x_k) \subseteq F$ has its limit in F.
b. Use it to show $[0,1]$ is closed and $(0,1)$ is not.

Question 9

a. Prove that if $x_k \to L$, then every subsequence (x_{k_j}) converges to L.
b. *(Challenging.)* Prove the **Bolzano–Weierstrass theorem**: every bounded sequence in $\mathbb{R}$ has a convergent subsequence. *Hint: first show every real sequence has a monotone subsequence, then apply the Monotone Convergence Theorem.*

Question 10

Let $f : X \to Y$ and $A, B \subseteq Y$. Prove the preimage identities:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B), \quad f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B), \quad f^{-1}(A^c) = \big(f^{-1}(A)\big)^c.$$

Question 11

Prove that the ε-δ and sequential definitions of continuity at a point are equivalent: for $f : X \to \mathbb{R}^m$ and $x_0 \in X$,

$$\big(\forall \varepsilon > 0\ \exists \delta > 0 :\ \|x - x_0\| < \delta \Rightarrow \|f(x) - f(x_0)\| < \varepsilon\big) \iff \big(x_k \to x_0 \Rightarrow f(x_k) \to f(x_0)\big).$$

Question 12

Let $f, g : X \to \mathbb{R}$ be continuous at x_0, and let $c \in \mathbb{R}$.

a. Using the sequential characterisation of continuity, prove that $f + g$, cf, and fg are continuous at x_0, and that f/g is continuous at x_0 provided $g(x_0) \neq 0$.
b. Prove that a composition of continuous functions is continuous: if $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Question 13

(Challenging.)

a. Prove the **Extreme Value Theorem**: if $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum on K.
b. Prove the **Intermediate Value Theorem**: if $f : [a, b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$.

ECON 600 tutorial outline

Advanced Microeconomic Theory I (PhD core), Vancouver School of Economics, UBC

William Duan, teaching assistant

A baseline plan for the weekly tutorials, refined over Fall 2025 and Fall 2026 and written from my notes on each session.[1] Each year follows the same sequence, adjusted to the pace of that year's lectures.

Purpose of the tutorials

Tutorials complement the lectures rather than repeat them. Each session works through a small number of problems, mostly from the problem sets and past exams, and uses them to make one or two ideas from lecture usable. Where the slides already cover a derivation well, the session supplies the interpretation instead.

Objectives

By the end of the term, students should have the following.

1. **Mathematical literacy.** The ability to parse a theorem, including what its premise and conclusion are and what it does not say, and to write simple proofs of their own.
2. **Command of the core material.** Enough understanding of the course content to pass the PhD comprehensive exam in microeconomic theory, given suitable effort on review.
3. **A picture of the whole.** A rough sense of how microeconomic theory builds from individual choice up to general equilibrium.

Schedule

The term has twelve weekly sessions. A holiday occasionally costs one, and make-up or exam review sessions balance it out.

Week 1 · Choice and the standard of proof. Rational preferences, choice rules and the Weak Axiom of Revealed Preference, ending with a first complete proof at the level of rigour the course expects.

Week 2 · Continuity and utility representation. The continuity axiom, how to show a preference is not continuous, and the theorem that a rational, continuous preference has a continuous utility representation.

Week 3 · The consumer's problem. Existence of a solution, Walras' law, and the convexity and uniqueness of demand, with attention to which hypothesis does the work in each result.

Week 4 · Proving from structure. A methods session. Results about demand under quasilinear and homogeneous utility, and a comparison of two taxes, each proved directly from the structure of the problem rather than by computing demand.

Week 5 · Duality and welfare. How Marshallian demand, indirect utility, the expenditure function and

[1] The Fall 2025 sessions are reconstructed from notes that Obleo Demandre took in the tutorials and kindly shared with me.

Hicksian demand determine one another, why consumer surplus is not a reliable welfare measure, and Afriat's theorem on rationalizing finitely many choices.

Week 6 · Testing rationality on data. Turning the generalized axiom of revealed preference into an algorithm and running it on choices from a laboratory experiment.

Week 7 · Choice over time. Present-biased preferences, commitment, and naive against sophisticated planners.

Week 8 · Choice under risk. The probability simplex, the continuity and independence axioms, and how to build counterexamples from a diagram.

Week 9 · Risk attitudes. The Allais paradox, risk aversion in portfolio choice, and precautionary saving.

Week 10 · Subjective expected utility. States, acts and subjective beliefs, and what choices between bets reveal about those beliefs.

Week 11 · Reference dependence and the firm. Personal equilibrium, the firm's problem and returns to scale, and monotone comparative statics.

Week 12 · General equilibrium. Walrasian equilibrium in economies with production, and how linear technologies pin down prices.

Year notes

Fall 2025. Eleven sessions. Week 1 reviewed proof techniques and basic real analysis. The representation theorem in week 2 was proved in full. Testing rationality on data was introduced briefly within week 5 rather than given its own session, so the later weeks each moved forward by one.

Fall 2026. In progress. I ran the Math Camp for this cohort, so week 1 went straight to choice and revealed preference instead of reviewing analysis. The representation proof in week 2 was previewed rather than replicated. Testing rationality on data had a full session, and choice over time was combined with the start of choice under risk.

ECON 600 · PROBLEM SET 3 · OPTIONAL CODING EXERCISE

Testing GARP on real choices

A step-by-step guide to checking whether each subject in the Data_LZ experiment makes choices that some utility function could explain. It follows the algorithm sketched in Tutorial 6.

How this exercise works

- Use any programming language you like.
- Use AI as much as you want, but not a frontier model. Pick a lighter one, such as the free Gemini app (Flash-Lite) or Claude Haiku.
- A weaker model forces you to explain the algorithm clearly. That skill is the point of the exercise. Honour code applies.

---

1 The data

Data_LZ has 82 subjects. Each subject faced the same 22 budget lines in two goods, x and y, and picked one bundle on each. One row is one choice. Subject IDs are labels, not a count from 1 to 82: they run from 105 to 2306 with gaps.

SUBJECT	OBSERVATION	X_CHOSEN	Y_CHOSEN	X_MAX	Y_MAX
105	1	25.88	25.88	100	34.92
105	2	10	40	20	80
105	3	30	30	61.57	58.5

`x_max` and `y_max` are where the budget line hits each axis. The line through them is

```
x / x_max + y / y_max = 1
```

So you can treat income as 1 and prices as `p_x = 1/x_max` and `p_y = 1/y_max`. Then the cost of any bundle (x, y) at the prices of budget i is

```
cost_i(x, y) = x / x_max_i + y / y_max_i
```

Every comparison below comes down to this cost.

---

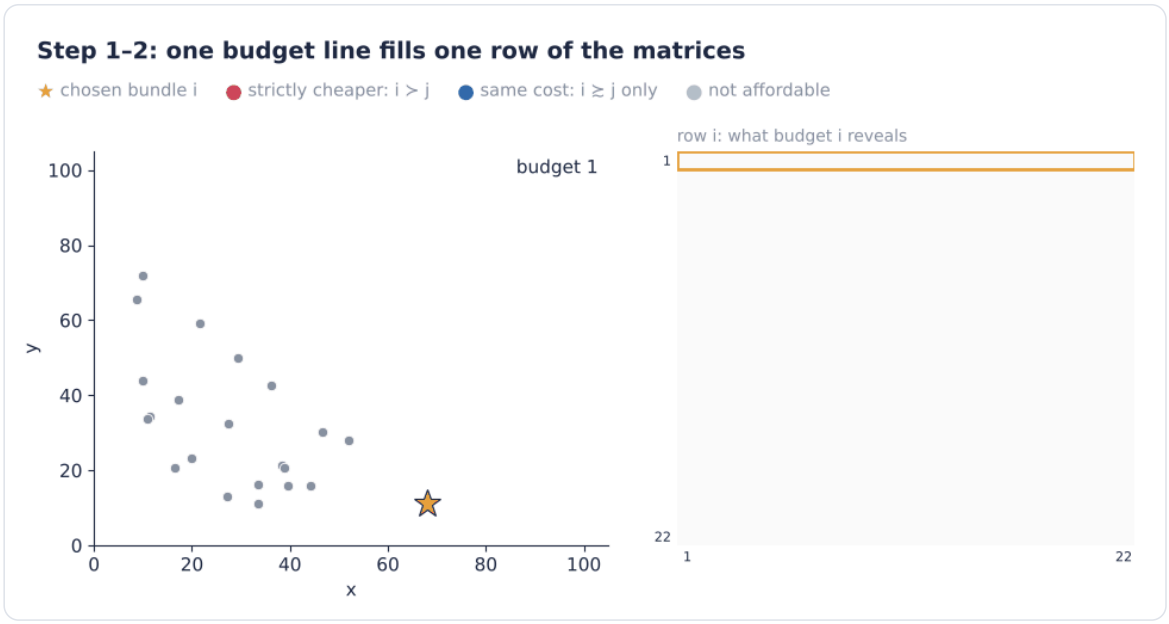
2 Comparing two choices

Take two observations i and j from the same subject. When the subject chose bundle i, was bundle j also affordable?

- **j cost strictly less than what was spent at i:** the subject passed up j while it was cheaper, so i is **strictly revealed preferred** to j, written $i \succ j$.
- **j cost exactly the same:** i is **weakly revealed preferred** to j, written $i \succsim j$.
- **j was too expensive:** we learn nothing. This does *not* mean $j \succsim i$. That would need its own check on budget j.

In symbols, with `spent_i = cost_i(bundle i)`:

```
i ≿ j   if   cost_i(bundle j) ≤ spent_i
i ≻ j   if   cost_i(bundle j) < spent_i
```



Each budget line answers the question for all 21 other bundles at once, which fills one row of the matrices. Subject 1106.

> **PITFALL: COMPARE WITH WHAT WAS SPENT, NOT WITH THE FULL BUDGET**
>
> The chosen bundles are rounded, so they are rarely exactly on the budget line. For subject 105, spending ranges from 0.9992 to 1.0001. In theory a subject spends the whole budget, so "cheaper than the full budget" and "cheaper than what was spent" are the same test. Once bundles are off the line, they differ. If you compare against the full budget (cost ≤ 1 with income normalised to 1) instead of against actual spending, rounding creates violations that aren't real.
>
> For example, subject 105 chose the same bundle (26.23, 26.23) on budgets 8 and 9. It costs 0.99992 under both, a hair inside each line. The budget-line rule gives $8 \succ 9$ *and* $9 \succ 8$, which is a GARP violation between two identical choices.
>
> Comparing against `spent_i` fixes this, and it also makes every bundle automatically weakly preferred to itself. That was the first fix suggested in the tutorial (set the diagonal to 1). The second fix, allowing some slack, comes back in the extensions.

---

3 Store every comparison

Run the comparison for every ordered pair (i, j) of one subject, which is 22 × 22 comparisons. Store the results in two matrices:

- **W**, where W[i][j] = 1 if $i \succsim j$
- **S**, where S[i][j] = 1 if $i \succ j$

Every strict link is also a weak one, so S[i][j] = 1 implies W[i][j] = 1. The diagonal of W is all 1s and the diagonal of S is all 0s. You could instead use one matrix with codes 0, 1 and 2, but two matrices keep the next step simpler.

Think of W as a directed graph: one node per observation, with an arrow i → j whenever $i \succsim j$. The rest of the algorithm searches this graph.

---

4 What a violation looks like

GARP fails when the choices chain back to where they started with at least one strict link along the way:

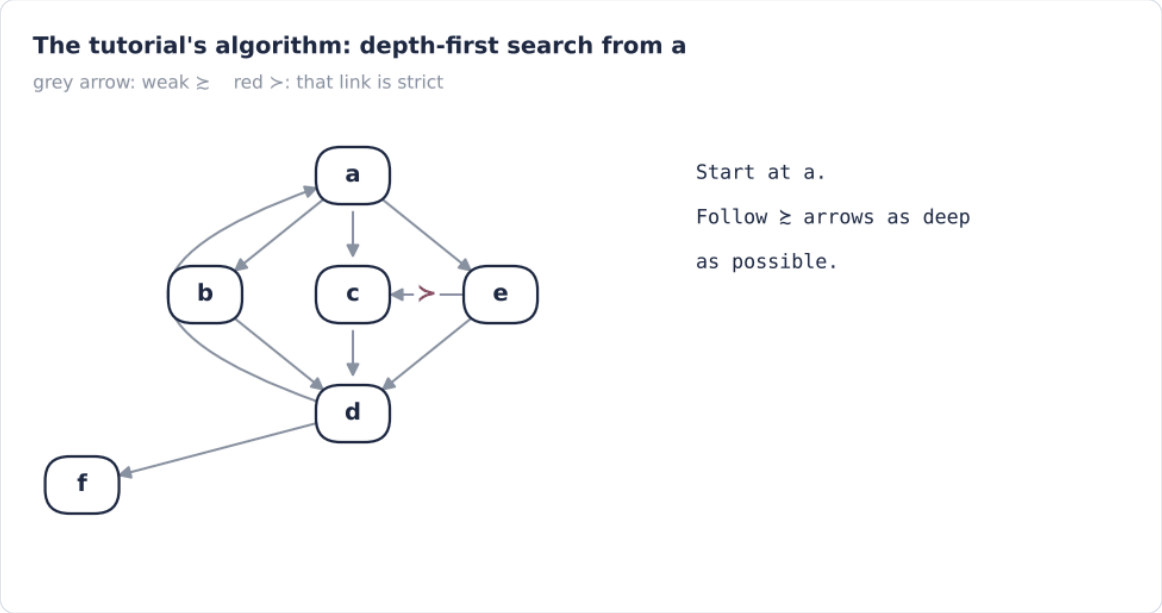
$$a \succsim b \succ c \succsim \dots \succsim a$$

In graph terms, this is a cycle of W arrows where at least one arrow is also in S. A cycle made only of weak links is fine: it just says the bundles are all equally good. So the task is to find every cycle and check each one for a strict link.

---

5 Find the cycles: search, check, merge

This is the algorithm from the tutorial. Depth-first search (DFS) explores the graph by following arrows as deep as possible, then backing up one step and trying the next branch. When the search returns to its starting node, it has found a cycle.



The tutorial example. Only the link e → c is strict. The search merges harmless cycles until it reaches the one that contains e ≻ c.

1. Pick a starting node s.
2. Run a DFS from s along the arrows of W. Each time the search comes back to s, you have a cycle.
3. Look up every link of that cycle in S. If any link is strict, stop: **GARP is violated**, and the cycle is your evidence.
4. If every link is weak, the nodes on the cycle are all indifferent. **Merge them into one node** that keeps every arrow in and out of its members. Then go back to step 2, starting from the merged node.
5. When the search from s finds no more cycles, s is cleared: no violation passes through it. Remove it from the graph and pick a new start.
6. If every node gets removed, the subject **satisfies GARP**.

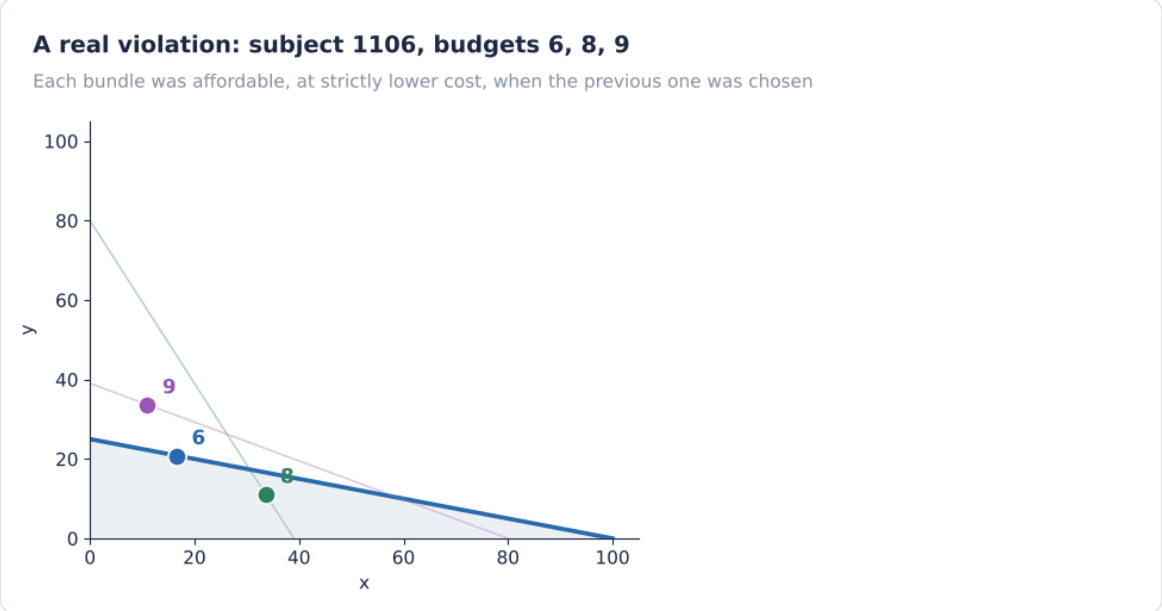
Two details are easy to miss:

- **Links into a merged node.** An arrow e → (abd) stands for several original links: e → a, e → b and e → d. It is strict if *any* of them is strict. The tutorial's reminder was to recheck all of them, for example a ≻ c or b ≻ c or d ≻ c.
- **Links inside a merged node.** Members of a merged node all lie on one weak cycle. So a strict link between two members, such as a ≻ d, is already a violation, even if the search reached d through b.

---

6 Run it on every subject

Loop over the 82 subjects, build W and S for each, and run step 5. Report which subjects pass, and for each one that fails, the cycle that breaks GARP.



A violation in the data, drawn on the budget lines. Subject 1106 strictly prefers 6 to 8, 8 to 9, and 9 to 6.

Revised October 2026 from the 2021 Problem Set 3 handout, following the Tutorial 6 sketch. Animations use subject data from Data_LZ.

Tutorial 7 Notes (2026–27)

William Duan

October 20, 2026

These notes document what was covered in the tutorial session, for the benefit of future TAs. They are not intended as study material for students.

Topics

The move from choice over goods to choice over lotteries. (i) The independence axiom, read off the probability simplex: what the simplex is and how to locate a lottery in it, why independence forces indifference curves to be straight and parallel, and how independence lets one "cancel" a common component of two lotteries. (ii) The continuity axiom for lotteries, and the one-directional use of its main consequence. (iii) A past exam question in which a lottery is valued by a weighted average of its mean and the top of its support: whether these preferences are rational, continuous and independent, worked through on the simplex. The session closed with advice on how much to write for discussion questions on an exam.

The formal statements of the axioms are on the lecture slides and in MWG Section 6.B. The figures at the end are reference sketches for a future TA preparing the board work.

Objective

The unifying theme of the expected-utility half is that the simplex is the working diagram for any question about axioms on lotteries, in the same way the two-good budget diagram was for consumer theory. Students who can draw the simplex fluently can see in a few seconds *where* a preference fails an axiom — on a curved indifference curve, at a non-parallel pair, at a jump on an edge — and then write down a counterexample built from exactly the lotteries they were looking at. The second half of that sentence matters as much as the first. The session made a point of drawing the picture first, saying explicitly that the picture is not a proof, and then turning it into one.

This also connects back to Tutorial 2, where continuity of preferences over goods was the subject and the lexicographic order the counterexample. The failure of continuity in the exam question has the same shape: a preference that is perfectly well-behaved in the interior and jumps on a boundary. Saying so explicitly lets students reuse an intuition they already have.

Comments

- *Reading the simplex.* I introduced the simplex as the set of probability vectors in three dimensions, the triangle cut out of the positive orthant by the plane where the probabilities sum to one, and then laid it flat. Each vertex is a sure outcome, and a lottery's probability on an outcome is its distance from the opposite edge, scaled so that the vertex is at distance one. I justified the last claim with a similar-triangles argument on the three-dimensional picture and then had the class locate a specific lottery (probabilities one quarter, one third and five twelfths) by drawing the three lines parallel to the edges.

  The similar-triangles derivation is the most compressible part of the session: the claim can be asserted, and the locating exercise is what students actually need to be able to do under exam conditions. The one-line version to leave them with is that "the probability of an outcome is how far you are from the edge where that outcome is impossible."

- *Independence: straight and parallel.* Stated as: indifference curves in the simplex are straight lines, and they are parallel. I gave the two halves separately, since they use independence differently.

  Straight lines: if two lotteries are indifferent, independence makes every mixture of them indifferent to both, so the indifference set through them contains the segment joining them. A curved indifference curve would require some point on that segment to be off the curve, which independence rules out. I drew a curved curve through two indifferent points and asked what independence says about the points on the segment between them, which made the contradiction visible immediately.

  Parallel: take two lotteries with one strictly preferred, and mix both with the same third lottery using the same weight. The two mixtures sit on a segment that is a scaled copy of the original segment, shrunk toward the third lottery, so by similar triangles it is parallel to the original. Independence says the ranking of the mixtures is the ranking of the originals. If indifference lines were not parallel, then for a weight small enough the two mixtures would land on opposite sides of some indifference line in the wrong order. The picture to draw is a fan of non-parallel lines with the segment shrinking toward the third lottery until its endpoints straddle a line the wrong way round.

  One caveat to keep in mind: I stated the result as an "if and only if," which is fine in the three-outcome picture for a complete and transitive preference, but students should not take the converse as a proof strategy outside the simplex. On an exam, independence is verified or refuted from the definition.

- *Cancelling a common component.* I used independence to simplify a comparison: a preference between the lottery with probabilities one quarter, one third and five twelfths and the uniform lottery, which I then presented as "subtracting" half the uniform lottery from both sides and rescaling. The arithmetic is right — the first lottery is an equal mixture of the uniform lottery and the lottery with probabilities one sixth, one third and one half — but the subtract-and-rescale framing passes through objects that are not lotteries, since the intermediate vectors no longer sum to one. A future TA should present it the other way round: write each lottery as a mixture with a common component, then apply the axiom. That is exactly what the definition licenses, and it avoids giving students the idea that lotteries can be subtracted.

- *Continuity and its consequence.* Stated in the MWG form: for any three lotteries, the set of mixing weights at which a mixture of the first two is at least as good as the third is closed, and so is the set at which it is at most as good. The consequence I emphasised is that if the third lottery lies between the first two in the ranking, some mixture is exactly indifferent to it.

  The point that needs saying out loud is that this consequence can only be used in one direction. Showing that no indifferent mixture exists proves that continuity fails. Exhibiting an indifferent mixture proves nothing, since the axiom is a statement about closedness for *every* triple, not existence for one. Proving continuity by finding an indifferent mixture is the natural mistake, and it is worth naming it as the failure mode before students make it.

  A related trap, which the board work brushed against: the axiom is not the same as closed contour sets, so a non-closed upper contour set does not by itself refute it. See the bonus note below.

- *The exam question.* The preference values a lottery at a weighted average of its mean and the top of its support. I wrote it in class with an unnormalised weight (the mean plus a multiple of the top outcome) and a specific weight of one tenth; the two forms represent the same family of preferences after rescaling.

  Rationality is immediate: any real-valued representation gives a complete and transitive order. Continuity and independence were done on a simplex with outcomes 100, 10 and 0. In the interior of the simplex, every lottery puts positive probability on 100, so the top of the support is constant and indifference curves are the parallel lines of the expected value. On the edge opposite the 100 vertex the top of the support drops discontinuously, so the edge is a different regime. I first gave the argument as a picture — an indifference line from an interior lottery reaches the bottom edge at a point drawn as an open circle, because that point has lost its upside and is strictly worse — and then said explicitly that the picture is not a proof, and that the task is to turn it into a counterexample with specific lotteries. The independence failure has the same source. Take a lottery that puts a sliver of probability on 100, so that it already collects the bonus for the top outcome, and a bottom-edge lottery with a higher mean that is ranked below it only because it lacks the bonus. Mixing both with the 100 vertex gives the second lottery the bonus too, the ranking reverts to expected value, and it reverses. A future TA should be careful here: mixing two bottom-edge lotteries with the 100 vertex does *not* produce a reversal, since both are ranked by their means before and after, and that is the configuration one is tempted to draw first.

  For continuity, a construction using sure lotteries over 0 and 10 is simpler than the one I gave in class. The question also asks whether the preference has an expected-utility representation. It does not, since independence fails, and students should be told to state that explicitly rather than leave it implied.

- *How much to write.* I ended with advice for the exam: discussion and interpretation questions want a good-enough answer of about three lines, not an essay. That advice is worth giving every year, before the midterm, and worth tying to the "what does the parameter represent" part of this question as an example of the right length.

Practical notes

On ordering: the simplex has to come before both axioms, since everything afterwards is drawn on it; and the exam question goes last because it uses both axioms and the simplex at once. If time is short, cut the similar-triangles derivation of the simplex coordinates and the cancellation example, in that order. The parallel-lines picture and the exam question are what justify the session.

Bonus note: mixture continuity versus closed contour sets

Not covered in the session, but a question a future TA should be ready for, since the board work came close to it. Is the continuity axiom for lotteries — closedness of the sets of mixing weights at which a mixture of two lotteries is at least as good as, or at most as good as, a third — equivalent to the upper and lower contour sets being closed in the simplex? It is not.

One direction holds. The map from a mixing weight to the corresponding mixture is continuous, and the axiom's sets of weights are exactly the preimages of the contour sets under it, so closed contour sets imply the axiom.

The converse fails, because the axiom only constrains the preference along line segments, and a function can be continuous along every line without being continuous. The standard example is $f(x,y) = x^2y/(x^4+y^2)$ with $f(0,0) = 0$. Restricted to any line it is continuous, but along the parabola $y = x^2$ it is constant at one half while the parabola runs into the origin, where $f = 0$. Place it around an interior point of the simplex and it represents a complete and transitive preference that satisfies the axiom, while its upper contour set at level one half is not closed.

The gap disappears once independence is added. Independence and the axiom together give an expected-utility representation, which is linear in probabilities and therefore has closed contour sets. So for preferences that satisfy independence the two notions coincide, and the distinction only matters when checking a preference that may fail independence — which is exactly the situation in the exam question.

The practical consequence for students is that exhibiting a non-closed upper contour set is not, on its own, a refutation of the axiom; the failure has to be shown along a segment of mixtures. In the exam question this is easy, because the offending sequence can be taken along a single indifference line ending at the bottom edge (the left panel of Figure 4). If students write the argument as "upper contour set not closed" on an exam, it should be tightened to name the segment.

Reference figures

Sketches of the board work, drawn to scale so that a future TA can check a picture before putting it up. Throughout, the simplex is equilateral, so a lottery's probability on an outcome is its distance from the opposite edge as a fraction of the triangle's height.

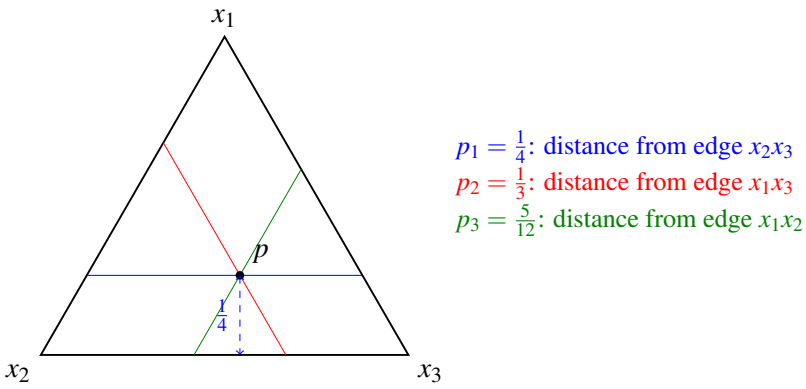


Figure 1: Locating the lottery $p = (\frac{1}{4}, \frac{1}{3}, \frac{5}{12})$. Each coloured line is the set of lotteries sharing one coordinate with p, parallel to the edge on which that outcome is impossible; the three meet at p.

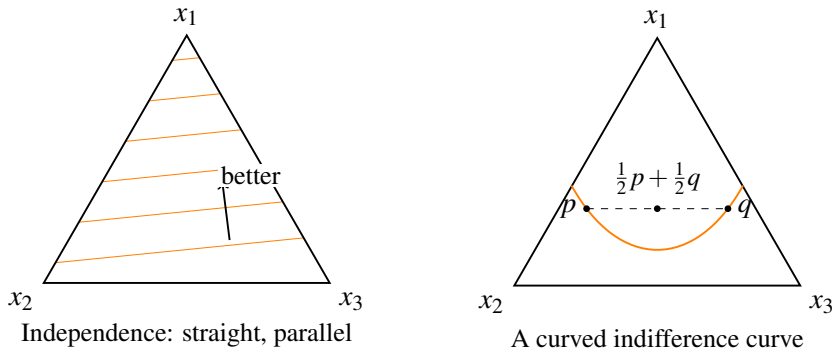


Figure 2: Left: under independence, indifference curves are parallel straight lines. Right: if $p \sim q$, independence gives $\frac{1}{2}p + \frac{1}{2}q \sim p$, but the midpoint lies off the curve through p and q, a contradiction.

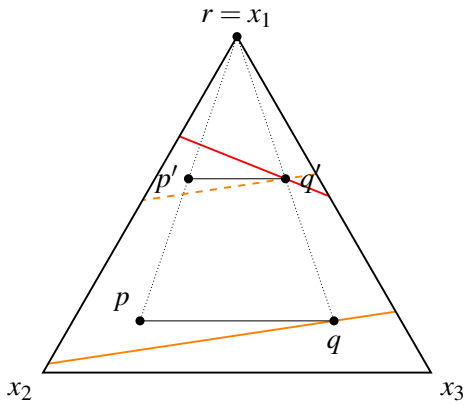


Figure 3: Why parallel. With $p' = \alpha p + (1-\alpha)r$ and $q' = \alpha q + (1-\alpha)r$ (here $\alpha = \frac{1}{2}$), the segment $p'q'$ is a scaled copy of pq, so it is parallel to it. The solid orange line is the indifference line through q, with p above it, so $p \succ q$. If the indifference line through q' were the red one, p' would lie below it and $q' \succ p'$, contradicting independence. The parallel dashed line keeps p' above it.

For the exam question the vertices are the sure outcomes 100 (top), 10 (bottom left) and 0 (bottom right). Every lottery off the bottom edge has 100 in its support, so there the value is $(1-\beta)$ times the mean plus the constant 100β, and the indifference curves are the lines of constant mean. These are almost horizontal, since the mean rises steeply toward the top vertex. On the bottom edge the top of the support falls to 10 (or to 0 at the bottom-right vertex), and the value drops.

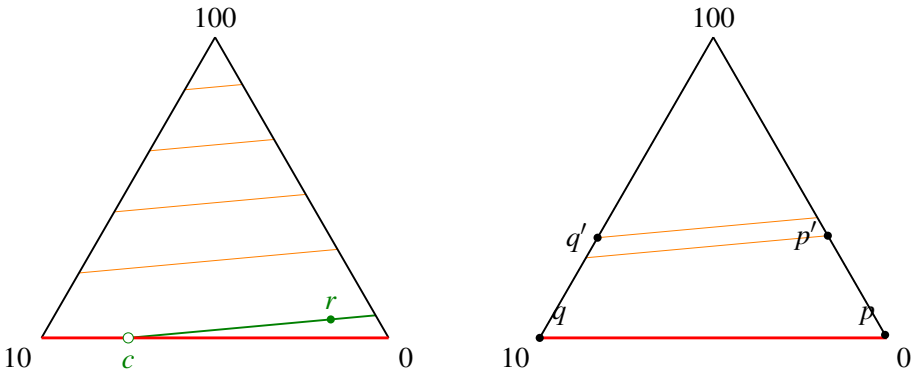


Figure 4: The exam question's preference; the red edge is where 100 drops out of the support. Left: the green indifference line through r meets the bottom edge at c, but c has lost the bonus for the top outcome and is strictly worse than r. So along the segment of mixtures from c to r, the set of weights giving something at least as good as r is half-open. This is a failure along a segment, which is what the axiom requires. Right: p puts probability 0.01 on 100 and the rest on 0; q is 10 for sure. With $\beta = \frac{1}{2}$, p is worth 50.5 and q is worth 10, so $p \succ q$ purely through the bonus. Mixing both with sure 100 at weight $\frac{1}{3}$ gives q' the bonus too; the means are now 34 for p' and 40 for q', so q' lies on a higher indifference line and the ranking reverses.

Discussion Session Week 2

William Duan

ECON 101, Fall 2020

1 Q1

Suppose Dontos derives the following enjoyment from drinking beer: What is

Number of Flagons	Total Enjoyment (measured in copper stars)	Marginal Benefit
1	20	
2	36	
3	48	
4	56	
5	60	
6	60	
7	56	

his marginal utility per flagon of beer? If one flagon of beer costs 4 copper stars, how many flagons would Dontos drink? What if it only costs 0.5 copper stars (also known as a groat)?

2 Q2

Assuming Dontos marginal benefit from beer is a linear function of the quantity of beer consumed, derive his demand curve.

3 Q3

If a tavern called the Green Dragon can supply beer at the following cost schedule: What is his marginal cost per flagon of beer? If one flagon of beer is priced at 4 copper stars, how many flagons would the tavern keeper sell?

4 Q4

Assuming marginal cost of producing beer is a linear function of the quantity of beer produced, derive the supply curve of the Green Dragon.

Number of Flagons	Total Cost (measured in copper stars)	Marginal Cost
10	1	
20	3	
30	6	
40	10	
50	15	
60	21	
70	28	

5 Q5

Suppose there are 100 taverns in Flea Bottom all with the same marginal cost as the Green Dragon, and every night, 1000 people (with exactly the same drinking habit as Dontos) frequent these taverns. What is the market demand curve and market supply curve of beer? What is the equilibrium quantity of beer consumed and what is the price?

6 Q6

Suppose due to a massive rise in religious fervour, demand for beer falls, what happens to the price and quantity of beer in the city? What if, because winter is coming and wheat is in short supply, the cost of making beer increases. What happens to the price and quantity of the beer? Describe a scenario that would cause price to rise but quantity consumed to remain unchanged.

THE UNIVERSITY OF BRITISH COLUMBIA

2025-26 Winter Term 1 UBC-V Individual TA Reports for ECON_V 600-L1A - Microeconomics I (Bufang Duan)

Project Title: **2025-26 Winter Term 1 UBC-V TA SEI Surveys**

Course Audience: **22**
Responses Received: **16**
Response Ratio: **73%**

Report Comments

Recommended Minimum Response Rates

Class Size	Recommended Minimum Response Rates based on 80% confidence & ± 10% margin
< 10	75%
11 - 19	65%
20 - 34	55%
35 - 49	40%
50 - 74	35%
75 - 99	25%
100 - 149	20%
150 - 299	15%
300 - 499	10%
> 500	5%

Legend

N: Invited
n: Responded

Frequency Distribution
SD: Strongly Disagree
D: Disagree

UBC Student Experience of Instruction

N: Neutral
A: Agree
SA: Strongly Agree

Creation Date: **Wednesday, January 28, 2026**

blue®

TA Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
The TA was helpful when I requested course-related assistance.	22	16	0	0	0	0	14	2	5.0	0.0
The TA was well-prepared for their duties.	22	16	0	0	0	2	13	1	4.9	0.1
The TA was readily available to me either through office hours or by appointment.	22	16	0	0	0	0	14	2	5.0	0.0
The TA communicated at an appropriate level for me.	22	16	0	1	0	3	11	1	4.8	0.3
The TA exhibited interest in the subject matter.	22	16	0	0	0	0	15	1	5.0	0.0
The TA showed knowledge of the subject matter.	22	16	0	0	0	0	15	1	5.0	0.0
The TA presented information clearly.	22	16	0	0	0	3	12	1	4.9	0.2
The TA was effective at stimulating interest in the subject matter.	22	16	0	0	0	4	11	1	4.8	0.2
The TA was receptive to a variety of perspectives and ideas.	22	16	0	0	1	2	11	2	4.9	0.2
The TA treated me and other students with equal respect.	22	16	0	0	1	2	12	1	4.9	0.2
The TA's comments on my written work were helpful.	22	16	0	0	0	2	9	5	4.9	0.1
The TA encouraged intelligent and independent thought.	22	16	0	0	1	3	11	1	4.8	0.3
The TA's evaluation of my work was fair and reasonable.	22	16	0	1	0	2	11	2	4.9	0.3
The TA made a very positive contribution to this course.	22	16	0	0	0	1	14	1	5.0	0.1

UBC Student Experience of Instruction

Question	%Favourable
The TA was helpful when I requested course-related assistance.	100%
The TA was well-prepared for their duties.	100%
The TA was readily available to me either through office hours or by appointment.	100%
The TA communicated at an appropriate level for me.	93%
The TA exhibited interest in the subject matter.	100%
The TA showed knowledge of the subject matter.	100%
The TA presented information clearly.	100%
The TA was effective at stimulating interest in the subject matter.	100%
The TA was receptive to a variety of perspectives and ideas.	93%
The TA treated me and other students with equal respect.	93%
The TA's comments on my written work were helpful.	100%
The TA encouraged intelligent and independent thought.	93%
The TA's evaluation of my work was fair and reasonable.	93%
The TA made a very positive contribution to this course.	100%

How would you rate the overall contribution of the TA to the course?

N	n	Very poor	Poor	Neutral	Good	Very Good	N/A	IM	DI
22	16	0	0	0	1	14	1	5.0	0.1

Question	%Favourable
How would you rate the overall contribution of the TA to the course?	100%

Of the sessions in which the Teaching Assistant was involved, how often did you attend?

Of the sessions in which the Teaching Assistant was involved, how often did you attend?

Options	Count	Percentage
None	1	6%
Less than 25%	0	0%
25-50%	1	6%
51-75%	0	0%
Greater than 75%	14	88%

UBC Student Experience of Instruction

Which, if any, of the following did your Teaching Assistant do? (Please mark each element that is appropriate)

Options	Count
Led discussion-tutorial groups	10
Held office hours	15
marked examination(s)	15
marked essay(s)-assignment(s)	11
gave classroom lectures	5
conducted labs	1
responded to email	9
Respondent(s)	15

Considering everything, how would you rate this course?

N	n	Very poor	Poor	Neutral	Good	Very good	N/A	IM	DI
22	16	0	0	0	3	12	1	4.9	0.2

Question	%Favourable
Considering everything, how would you rate this course?	100%

Please comment on any aspects, positive or negative, of your teaching assistant's (TA's) teaching, attitudes to students, class atmosphere, or any other matters affecting the quality of instruction that you consider worthy of note.

Comments
na
He is very helpful to understand how to solve exercises. He always was very open to solve our doubts. He sometimes extended OHs to ask all our questions, which I really appreciate.
William is a fantastic TA. He is very committed to the students (e.g., spending lots of extra time helping students and answering questions beyond what is required of him). He has a mastery of the material and emphasizes relevant concepts.
TA is super helpful and everyone agreed. He is able to tailor the level of instruction to everyone's needs. He is able to explain plainly, from many different aspects, when I asked a entry–level question. And he is also able to answer all my questions that are a little bit beyond course requirements, and some of my research–related questions that is even not related to the course materials. He is very enthusiastic, willing to help, and his office hour is almost always lasting longer than required in order to answer everyone's question. He is also super knowledgeable; I would reasonably expect that he is able to answer all of my questions, and would reach back with accurate and clear answer really soon if not on the spot. It is super great to have him as the TA for a difficult class like 600. Everyone appreciated his efforts.
William is clearly very passionate and knowledgeable about the subject and conveyed concepts clearly.

UBC Student Experience of Instruction

Please comment on any aspects, positive or negative, of the format and content of the course as they may have affected the teaching assistant's performance.

Comments
na
Having William as TA greatly enhanced this course. I have no major complaints.